CLASSIFICATION

COUNTING THE COST – PART I

Fabio Stella

Associate Professor

c/o Department of Informatics, Systems and Communication

University of Milano Bicocca

Counting the Cost

The following concept will be introduced:

✓ **Cost Matrix**

Counting the Cost

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	n	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	y	7	1.78	OH	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	n	6	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA	510	355-9993
147	0	157	n	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173

After having concluded the **Data Mining** analysis you write a **report** on the **Churn Problem** and enclose it to an e-mail message you send to you friend.

After having read the report your friend calls you on the phone and tells that **to limit churning** the Telecom Company plans **a promotional action which can be offered to no more than 500 customers**.

COUNTING THE COST

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You partition the **CUSTOMERS CALLS DATABASE** to obtain:

- ✓ **TRAINING DATA SET**
- ✓ **TESTING DATA SET**

You plan a **CLASSIFICATION STUDY** using the Training Data Set to develop **DIFFERENT CLASSIFICATION MODELS TO BE COMPARED ON THE TEST DATA SET**.

You select the **CLASSIFICATION MODEL (M)** as the **OPTIMAL INDUCER** to be used to make the decision to **APPLY OR NOT THE PROMOTIONAL ACTION TO A GIVEN CUSTOMER**.

To persuade your friend that the **CLASSIFICATION MODEL (M) IS EFFECTIVE** and that it **ACHIEVES A BETTER PERFORMANCE THAN THE STANDARD MAILOUT PROCEDURE (SMP)** you ask your friend to provide the inducer of the Standard Mailout Procedure.

You apply the **STANDARD MAILOUT PROCEDURE (SMP)** and the **CLASSIFICATION MODEL (M)** to the **TEST DATA SET** to obtain their **CONFUSION MATRICES**:

SMP	PREDICTED CLASS		
		-1	+1
ACTUAL CLASS	-1	830	111
	+1	45	114

M	PREDICTED CLASS		
		-1	+1
ACTUAL CLASS	-1	931	10
	+1	57	102

You show your friend that the **CLASSIFICATION MODEL (M) ACHIEVES A BETTER PERFORMANCE THAN THE STANDARD MAILOUT PROCEDURE (SMP)**, the **ACCURACY** values are computed using the independent **TEST DATA SET**.

Counting the Cost

You apply the **Standard Mailout Procedure (SMP)** and the **Classification Model (M)** to the **Test Data Set** to obtain their **Confusion Matrices**:

SMP	PREDICTED CLASS	
ACTUAL CLASS	**-1**	**+1**
-1	830	111
+1	45	114

Accuracy = 0.858

M	PREDICTED CLASS	
ACTUAL CLASS	**-1**	**+1**
-1	931	10
+1	57	102

Accuracy = 0.939

You show your friend that the **Classification Model (M) achieves a better performance than the Standard Mailout Procedure (SMP)**, the **Accuracy** values are computed using the independent **Test Data Set**.

Counting the Cost

You apply the **Standard Mailout Procedure (SMP)** and the **Classification Model (M)** to the **Test Data Set** to obtain their **Confusion Matrices**:

SMP	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	830	111
	+1	45	114

Accuracy = 0.858

M	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	931	10
	+1	57	102

Accuracy = 0.939

However, your friend informs you that **THE COST OF CLASSIFYING A CHURNER CUSTOMER AS A NON CHURNER IS FAR GREATER THAN THE COST OF CLASSIFYING A NON CHURNER CUSTOMER AS A CHURNER.**

Counting the Cost

You apply the **Standard Mailout Procedure (SMP)** and the **Classification Model (M)** to the **Test Data Set** to obtain their **Confusion Matrices**:

SMP	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	830	111
	+1	45	114

Accuracy = 0.858

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		-1	**+1**
ACTUAL CLASS	**-1**	931	10
	+1	57	102

Accuracy = 0.939

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	0	1
	+1	100	-1

However, your friend informs you that **THE COST OF CLASSIFYING A CHURNER CUSTOMER AS A NON CHURNER IS FAR GREATER THAN THE COST OF CLASSIFYING A NON CHURNER CUSTOMER AS A CHURNER.**

Counting the Cost

You apply the **Standard Mailout Procedure (SMP)** and the **Classification Model (M)** to the **Test Data Set** to obtain their **Confusion Matrices**:

SMP	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	830	111
	+1	45	114

Accuracy = 0.858

Cost = 4,497

M	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	931	10
	+1	57	102

Accuracy = 0.939

Cost = 5,608

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	0	1
	+1	100	-1

However, your friend informs you that **the cost of classifying a churner customer as a non churner is far greater than the cost of classifying a non churner customer as a churner.**

Counting the Cost

You apply the **Standard Mailout Procedure (SMP)** and the **Classification Model (M)** to the **Test Data Set** to obtain their **Confusion Matrices**:

SMP	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	830	111
	+1	45	114

Accuracy = 0.858

Cost = 4,497

M	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	931	10
	+1	57	102

Accuracy = 0.939

Cost = 5,608

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	0	1
	+1	100	-1

Accuracy makes us to prefer the Classification Model (M)

COUNTING THE COST

You apply the **STANDARD MAILOUT PROCEDURE (SMP)** and the **CLASSIFICATION MODEL (M)** to the **TEST DATA SET** to obtain their **CONFUSION MATRICES**:

SMP	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	830	111
	+1	45	114

ACCURACY = 0.858

COST = 4,497

M	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	931	10
	+1	57	102

ACCURACY = 0.939

COST = 5,608

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	0	1
	+1	100	-1

COST MAKES US TO PREFER THE STANDARD MAILOUT PROCEDURE (SMP)

Counting the Cost

The **cost incurred by a Classification Model** is computed as follows:

	PREDICTED CLASS		
ACTUAL CLASS		**-1**	**+1**
	-1	TN	FP
	+1	FN	TP

	PREDICTED CLASS		
ACTUAL CLASS		**-1**	**+1**
	-1	C_{--}	C_{-+}
	+1	C_{+-}	C_{++}

$$\text{Cost} = C_{--} \cdot TN + C_{-+} \cdot FP + C_{+-} \cdot FN + C_{++} \cdot TP$$

COUNTING THE COST

The **COST INCURRED BY A CLASSIFICATION MODEL** is computed as follows:

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	TN	FP
	+1	FN	TP

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	C_{--}	C_{-+}
	+1	C_{+-}	C_{++}

$$\text{Cost} = C_{--} \cdot TN + C_{-+} \cdot FP + C_{+-} \cdot FN + C_{++} \cdot TP$$

$$\text{Cost} = 0 \cdot 830 + 1 \cdot 111 + 100 \cdot 45 + (-1) \cdot 114 = 4{,}497$$

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	830	111
	+1	45	114

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	0	1
	+1	100	-1

Counting the Cost

Notice that when **costs coefficients are symmetric** the **Cost is proportional to Accuracy**.

	PREDICTED CLASS		
		-1	+1
ACTUAL CLASS	-1	TN	FP
	+1	FN	TP

	PREDICTED CLASS		
		-1	+1
ACTUAL CLASS	-1	p	q
	+1	q	p

$$N = TP + FN + FP + TN$$

COUNTING THE COST

Notice that when **COSTS COEFFICIENTS ARE SYMMETRIC** the **COST IS PROPORTIONAL TO ACCURACY**.

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	**TN**	**FP**
	+1	**FN**	**TP**

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	**p**	**q**
	+1	**q**	**p**

N = **TP** + **FN** + **FP** + **TN**

ACCURACY = (**TP**+**TN**)/N

COUNTING THE COST

Notice that when **COSTS COEFFICIENTS ARE SYMMETRIC** the **COST IS PROPORTIONAL TO ACCURACY**.

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	**TN**	**FP**
	+1	**FN**	**TP**

	PREDICTED CLASS		
		-1	**+1**
ACTUAL CLASS	**-1**	**p**	**q**
	+1	**q**	**p**

N = **TP** + **FN** + **FP** + **TN**

ACCURACY = (**TP**+**TN**)/N

COST = **p** · (**TP**+**TN**) + **q** · (**FN**+**FP**)

COUNTING THE COST

Notice that when **COSTS COEFFICIENTS ARE SYMMETRIC** the **COST IS PROPORTIONAL TO ACCURACY.**

	PREDICTED CLASS		
ACTUAL CLASS		-1	+1
	-1	TN	FP
	+1	FN	TP

	PREDICTED CLASS		
ACTUAL CLASS		-1	+1
	-1	p	q
	+1	q	p

$$N = \mathbf{TP} + \mathbf{FN} + \mathbf{FP} + \mathbf{TN}$$

$$\textbf{ACCURACY} = (\mathbf{TP}+\mathbf{TN})/N$$

$$\begin{aligned}\textbf{COST} &= \mathbf{p} \cdot (\mathbf{TP}+\mathbf{TN}) + \mathbf{q} \cdot (\mathbf{FN}+\mathbf{FP}) \\ &= \mathbf{p} \cdot (\mathbf{TP}+\mathbf{TN}) + \mathbf{q} \cdot (N-\mathbf{TP}-\mathbf{TN})\end{aligned}$$

Counting the Cost

Notice that when **costs coefficients are symmetric** the **Cost is proportional to Accuracy**.

	PREDICTED CLASS		
ACTUAL CLASS		-1	+1
	-1	TN	FP
	+1	FN	TP

	PREDICTED CLASS		
ACTUAL CLASS		-1	+1
	-1	p	q
	+1	q	p

$$N = TP + FN + FP + TN$$

$$\text{Accuracy} = (TP+TN)/N$$

$$\begin{aligned}\text{Cost} &= p \cdot (TP+TN) + q \cdot (FN+FP) \\ &= p \cdot (TP+TN) + q \cdot (N-TP-TN) \\ &= q \cdot N - (q-p) \cdot (TP+TN)\end{aligned}$$

Notice that when **COSTS COEFFICIENTS ARE SYMMETRIC** the **COST IS PROPORTIONAL TO ACCURACY**.

	PREDICTED CLASS		
ACTUAL CLASS		-1	+1
	-1	TN	FP
	+1	FN	TP

	PREDICTED CLASS		
ACTUAL CLASS		-1	+1
	-1	p	q
	+1	q	p

$$N = \mathbf{TP} + \mathbf{FN} + \mathbf{FP} + \mathbf{TN}$$

$$\text{ACCURACY} = (\mathbf{TP}+\mathbf{TN})/N$$

$$\begin{aligned}
\text{COST} &= \mathbf{p} \cdot (\mathbf{TP}+\mathbf{TN}) + \mathbf{q} \cdot (\mathbf{FN}+\mathbf{FP}) \\
&= \mathbf{p} \cdot (\mathbf{TP}+\mathbf{TN}) + \mathbf{q} \cdot (N-\mathbf{TP}-\mathbf{TN}) \\
&= \mathbf{q} \cdot N - (\mathbf{q}-\mathbf{p}) \cdot (\mathbf{TP}+\mathbf{TN}) \\
&= N \cdot [\mathbf{q} - (\mathbf{q}-\mathbf{p}) \cdot \text{ACCURACY}]
\end{aligned}$$